IMO 2021 Shortlist, Problem A5

Let $n \geq 2$ be an integer and let $a_1, a_2, \ldots, a_n$ be positive real numbers with sum 1. Prove that

$$\sum_{k=1}^{n} \frac{a_k}{1-a_k}(a_1 + a_2 + \cdots + a_{k-1})^2 < \frac{1}{3}.$$



Formalization system

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theorem imo_shortlist_2021_a5
    (n : ℕ) (h₀ : 2 ≤ n) (a : ℕ → ℝ) (hapos : ∀ i, 0 < a i)
    (hasum : ∑ i in Finset.Icc 1 n, a i = 1) :
    ∑ k in Finset.Icc 1 n, a k / (1 - a k) * (∑ i in Finset.Icc 1 (k-1), a i) ^ 2 < 1 / 3
```